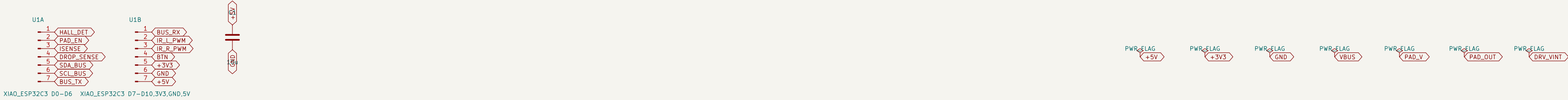
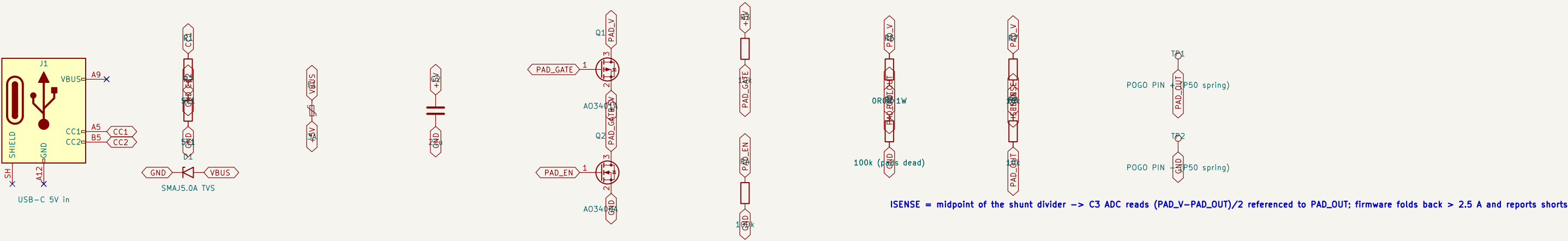


ORBIE V5 – DOCK BASE BOARD r0.1. Pads dead until a robot is present; feeder works without the robot; 70 x 54 mm, 2-layer.

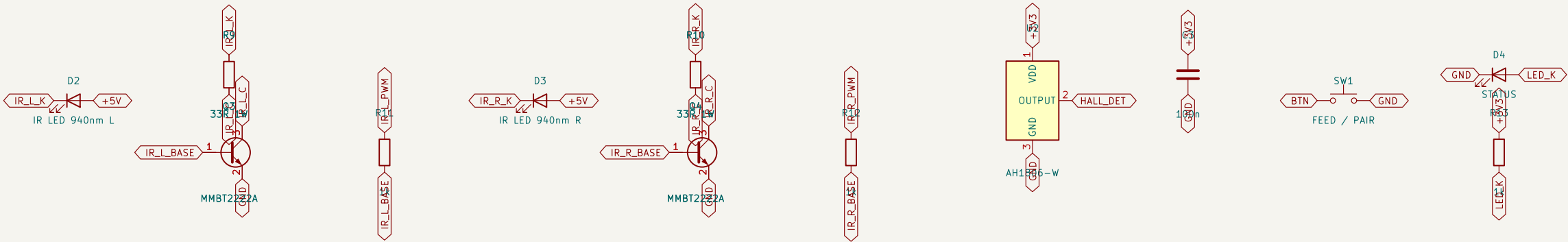
U1 XIAO ESP32-C3 – dock controller (Wi-Fi to app, schedules feeding without the robot, REQ-D3). D0 hall, D1 pad enable, D2 ISENSE (ADC), D3 drop sensor, D4/D5 bus I2C, D6/D7 UART to bus, D8 IR_L, D9 IR_R, D10 button



POWER IN USB-C 5V/3A (CC 5k1), TVS, 3A polyfuse. PADS: AO3401A high-side switch, 100k pull-down, 0.05R shunt + ADC fold-back; pads are DEAD until hall + handshake (REQ-D1)



HOMING 2x 940 nm IR LEDs (L/R lobes) at 38 kHz from the C3 via MMBT2222A, 33R from 5V. U2 hall AH1806: robot magnet = present. SW1 button: feed now / hold = pair



MODULES U3 DRV8833 drives the feeder paddle motor (A) and the water pump (B) through the stacking bus J2 (10p: 5V GND FEED_A FEED_B PUMP_A PUMP_B DROP PRESENT_F LEVEL PRESENT_W). Modules are dumb; motors live in the modules (REQ-D4)

